

Technical Skills

Languages	C#, Kotlin, Java, TypeScript, JavaScript
Libraries / Frameworks	Angular, React, Node, .NET, Web API, Entity Framework, Dapper
Databases	MongoDB, SQL Server, PostgreSQL, DynamoDB, CosmosDB
Systems / Platforms	GCP, AWS, Azure, Kubernetes, Docker, Helm

Work Experience

Oct '23 - Present	Software Engineer - TAINA Tech Ltd.TAINA Validator - Led the containerisation of a .NET 6 Web API and React project with Docker, substantially improving development workflows and deployment consistency. Expertly implemented Kubernetes in Amazon EKS for application orchestration, utilising Helm charts for streamlined deployment and management of services. This strategic integration fostered a scalable, resilient hosting solution on AWS, optimising operational efficiency and minimising infrastructure overhead - Transformed database migration processes by transitioning from manual scripts to implementing FluentMigrations, automating migrations to execute upon project startup. This innovation significantly streamlined database updates, enhancing consistency and reducing manual workload - Initiated and executed a comprehensive refactoring of the API project, focusing on the efficient management of service lifetimes. This pivotal effort significantly elevated the application's performance and streamlined maintenance processes - Identified a critical SQL Server connection leak in the first week at the company, causing severe performance slowdowns for customers using SQL Server - Prioritised the transition from end-to-end UI testing to integration testing using programmatic interfaces, significantly speeding up test and deployment processes while enhancing test robustness - Mentored junior team members through fortnightly sessions, focusing on clean code practices and encouraging them to extend their capabilities beyond their comfort zones. These mentorship activities aimed to foster professional growth, enhance code quality, and build a culture of continuous learning and improvement within the team
Oct '22 - Sep '22	Software Engineer - Dojo Ltd.Communications & Identity - Worked alongside a compact and nimble team to create and sustain a shared service platform, streamlining the process of crafting services for in-house teams. These services were implemented in .NET 7, Kubernetes, and Docker, utilising a CQRS architectural approach and employing MongoDB - Successfully implemented an event-driven architecture on Google Cloud Platform (GCP) using PubSub, enhancing the platform's scalability and responsiveness, using a Domain-Driven Design - Participated in the upgrade initiative for all services that leverage our service, enabling them to embrace the latest version of our library and PubSub event model, and successfully transitioning all services to .NET 7. This expedited the retirement of the legacy system, resulting in a rapid and efficient transition - Part of an initiative to redesign the core internal service, consolidating it into a centralised instance that served as the primary point of communication for all other services, eliminating the need for ephemeral environments - Participated in a project to develop an in-house Identity service to replace the legacy system. Contributed to architectural decisions, integrated Auth0, and coordinated project timelines - Conducted database migration operations to modernise legacy databases by transferring historical storage data, converting it into the updated data model, and seamlessly integrating it into the new database infrastructure, thereby ensuring the synchronisation and advancement of the new database system
Oct '21 - Aug '22	Software Engineer - HealthHero Ltd.MedSol PWA / SAPI - Migrated old monolithic front-end repository into a monorepo solution, and rewrote the various client implementations into individual PWAs written in React - Carried out a project to introduce an adequate test framework into a .NET API named SAPI which had little-to-no testing in place, resulting in a Unit Test coverage of above 80% and an Integration Test coverage of above 70%

- Contributed to a task force addressing critical bugs in a client API wrapper, leading to a 2-year contract renewal
- Contributed to the development of a specialised database migration tool tailored for Azure CosmosDB, enabling the anonymisation of personally identifiable information (PII) data to ensure compliance with GDPR regulations while facilitating data analysis capabilities for cross-functional teams

Jul '19 -
Oct '21

Junior Software Engineer - Argus Media Ltd.

Direct Workspaces

- Engaged in a greenfield project tasked with the creation of a highly customisable user workspace, designed to offer critical price benchmark data. Developed utilising Angular and .NET Core, the project incorporated a robust database architecture, leveraging both Amazon DynamoDB and SQL Server
- Implemented the Noda Time API to overcome the various design limitations in the .NET Time API
- Carried out a project to introduce a theming service to asynchronously load and cache selected custom themes for the user, resulting in clients opting to use non-default themes at a rate of 84%
- Attending monthly security advocacy meetings to identify areas of vulnerability within the code base and infrastructure DevOps pipeline
- Collaborated on a greenfield project using AWS API Gateway and serverless computing to implement company-built custom workspaces

Jun '18 -
Jun '19

QA Engineer - ASOS PLC

Bag API

- Implemented new features to improve the user experience on an Azure hosted C# Facebook Chatbot
- Developed and implemented a stored procedure to automatically remove test-user entries from CosmosDB during each deployment
- Introduced brand mapping functionality so end-users could specify specific brands within their searches, thus allowing users to see what they actually want to see
- Attended twice-weekly code craftsmanship classes run by the Lead Developer in order to make the transfer over to Software Engineer
- Completely refactored existing test suite left behind by prior Senior QA to make it more robust and mapped according to the Testing Pyramid. Resulted in reduced deployment times of around 40% and a less "flaky" test suite

May '17 -
Jun '18

Associate Software Developer in Test - Veritas LLC

Information Classifier

- Implemented new features on a Java microservice for data discovery and visualisation. An NLP solution implementing Stanford NLP, using a TDD and BDD approach to development
- Created an internal performance monitoring dashboard using Chart.js to monitor any degradation in the classification engine
- Writing Unit, Integration, and UI tests using Mockito, TestNG, and Protractor for the End-to-End tests
- Configured and remastered build & release pipelines with CI/CD using Maven, Jenkins and JIRA

Projects

Automated MMORPG Scripts

As a passionate gamer, my curiosity led me to explore the creation and execution of automated scripts in online games. During my exploration, I came across an open platform where I had the opportunity to engage with a deobfuscated version of the game Runescape, and interact with an open-source API developed for it. You can find the plugins related to this exploration on my GitHub page here: [harvey-plugins](#)

AWS Stock Checker

Created a straightforward, functional AWS script designed to monitor the availability of a specific item on a designated website and automatically notify the user via email when the item becomes available. Developed this project in response to a friend's request for a custom notifier solution when faced with the absence of built-in notification options

Artificial Neural Network for Intrusion Detection

Final year project to develop a Neural Network for aid in intrusion detection, using the NSL-KDD dataset. Trained the Network using a back-propagation method to detect intrusions by analysing packet attributes. Programmed and trained the network in MATLAB and resulted in an 87% detection rate. Paper available at [harveysingh.co.uk/NeuralNetwork.pdf](#), and files available on request

Education

2012-2017 | Bachelor of Science in **Computer Science**
Brunel University, London

Award: Second Upper Class (2:1)